

Rapid Prototyping & Evaluation of Crew Stations to Support Manned-Unmanned teaming

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ABSTRACT

Crew stations are one of the most important enablers of human-machine teaming on battlefields to support remote operations of robotic combat vehicles. Emerging technologies such as virtual and augmented reality offer opportunities to rapidly configure and test new crew interface concepts. However, what would be the appropriate fidelity and flexibility of virtual prototypes that produce user experience and performance comparable to the real-world? Based on Milgram and Kishino's reality-virtuality continuum [1], this project prototyped crew stations in (1) virtual environment (VE), (2) augmented virtuality (AV), and (3) augmented reality (AR) with varying fidelity and flexibility and evaluated user experience and performance as compared to the crew station prototyped in the (4) real environment (R). Preliminary results from our human-subject experiment show visual search time in AV and task completion time in AR are comparable to real-world conditions. Error rates, situation awareness, workload, and preference across AR, AV, and VR matched real-world. However, presence and usability were lower than real world, highlighting areas for improvement. These findings support AR, AV, and VR as promising tools for rapid prototyping, with future work targeting improved immersion, usability, and broader validation.

Keywords: Augmented Reality (AR), Augmented Virtuality (AV), Virtual Reality (VR), Crew Station Prototypes, Haptic Feedback, Human-Computer Interaction (HCI).

Index Terms: Operator Performance, Situation Awareness, Workload, Perceived Presence, System Usability, Flexible Prototyping, Human-Subject Experiment.

1 INTRODUCTION

The development and deployment of remote operations of military vehicles have become integral to modern defense strategies, particularly through manned-unmanned teaming (MUM-T). MUM-T allows human operators and autonomous systems to collaborate effectively, combining the decision-making abilities of a human crew with the precision and operational endurance of unmanned vehicles [2][3]. In this context, crew stations play a critical role as the interface for control and communication with manned and unmanned systems, enabling seamless operations in complex and dynamic environments. While significant advancements have been made in military vehicle technology, challenges persist in designing and evaluating crew stations that facilitate effective interaction between operators and unmanned systems. The development of crew station prototypes requires the ability to rapidly configure and test new interface concepts to ensure optimal user experience and operational efficiency [4].

The integration of head/helmet-mounted displays (HMDs) aim to enhance crew-system interaction. However, achieving a balance between fidelity (realism) and flexibility (adaptability) of these virtual prototypes remains a significant research challenge. These technologies offer immersive and realistic training environments, but their implementation must be carefully considered and studied.

2 OBJECTIVE

This project aims to study and evaluate prototyping of crew stations and interaction techniques to support human-machine teaming by achieving two primary research objectives (RO) and answer one research question (RQ).

RO1: Develop a platform & procedures for crew station prototyping in AR/VR to support remote operations of unmanned vehicles.

RO2: Validate the proposed platform and procedures via a human-subject experiment.

RQ: What are the appropriate fidelity and flexibility of virtual prototypes that produce user experience and performance comparable to the real world?

3 METHOD

We developed a configurable and interactable crew station prototyping platform across four distinct environments: (1) Real Environment, (2) Augmented Reality (AR), (3) Augmented Virtuality (AV), and (4) Virtual Environment (VR). The Real Environment (R) represented a fully functional crew station with working displays and controls, providing the highest fidelity and lowest flexibility. In the Augmented Reality (AR) environment, physical mockups of displays and controls were overlaid with virtual components, such as interactive displays, offering high fidelity but low flexibility due to physical constraints. The Augmented Virtuality (AV) setup used mockups made of cardboard and 3D-printed controls, combined with virtual overlays, creating medium fidelity and flexibility. Lastly, the Virtual Environment (VR) was fully simulated with no physical components, providing the lowest fidelity and the highest flexibility.

To validate the platform, a 4 x 1 within-subject experiment was conducted with 8 participants. Participants completed a human-machine teaming search and rescue task in all four crew station prototypes. The task required participants to control a robotic vehicle remotely, using a crew station interface to search for and identify target figures within a rescue area. The independent variable was the prototype (R, AR, AV, VR) and we observed the effect on user performance and user experience. Dependent variables included performance metrics such as task completion time, visual search time, error rate, and situation awareness (SART questionnaire) [5] and user experience metrics like presence (Slater, Usoh, and Steed questionnaire) [6], usability (System Usability Scale) [7], workload (Modified Cooper-Harper Scale) [8], and preference (rankings).

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4 RESULTS

The results indicate that visual search time for the AV prototype is comparable to the real-world condition. Similarly, the total task completion time for the AR prototype is comparable to the real-world condition. Furthermore, the error rates across all prototypes -AR, AV, and VR- are comparable to the real-world condition, indicating that none of the tested prototypes introduced additional errors during the visual search tasks.

For operator situation awareness, the AR, AV, and VR prototypes are comparable to the real-world condition, suggesting that participants maintained similar levels of awareness across all conditions. Regarding user experience, the presence ratings for prototypes in AR, AV, and VR were worse than the real-world condition, indicating a noticeable decline in immersion. However, system usability for the AV and VR prototypes was comparable to the real-world baseline, showing no significant usability challenges. Both workload and preference ratings for AR, AV, and VR prototypes were also comparable to the real-world condition, suggesting that participants experienced similar levels of mental effort and preference across all conditions.

5 CONCLUSION

Preliminary results from a human-subject experiment (n=8) indicate that haptic or tactile feedback enabled operator performance in AR and AV prototypes to be comparable to real-world conditions. Additionally, situation awareness, workload, and preference across all prototypes (AR, AV, VR) matched real-world benchmarks, demonstrating the effectiveness of the proposed platform in maintaining critical performance and cognitive outcomes. However, perceived presence and usability remain areas for improvement, as these were rated lower in other conditions compared to real-world conditions. These findings suggest that while the prototypes successfully replicate key operational metrics, enhancements are needed to improve user experience, particularly in presence and usability, to match real-world standards fully.

6 FUTURE WORK

Enhancements to prototypes will be explored to create a more immersive and realistic experience, aiming to bridge the gap between the prototypes and real-world conditions. Additionally, further iterations of the platform will emphasize refining visual and auditory fidelity to increase presence. The research team will also conduct experiments with a larger and more diverse participant pool to validate the generalizability of the findings. Investigations into adaptive interfaces and personalization of crew station configurations will be pursued to optimize operator performance and situation awareness across varying task demands. Lastly, longitudinal studies will assess the impact of prolonged exposure to AR, AV, and VR environments on user workload, fatigue, and overall system effectiveness in operational settings.

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